

Context-specific, directional, genetics-anchored drug-target nomination from genome-scale CD4+ T-cell CRISPRi Perturb-seq

Analysis report — computational target discovery

1. Summary

We re-analyzed a genome-scale CRISPRi Perturb-seq screen in primary human CD4+ T cells (Zhu, Dann, ... Pritchard & Marson 2025, bioRxiv 10.64898/2025.12.23.696273) to nominate drug targets for autoimmune and allergic disease. Our angle is deliberately differentiated from the preprint's own target list along three axes:

1. **Directionality** — we separate targets whose knockdown *lowers* a pathogenic pro-inflammatory program (**drivers** → **antagonize**) from those whose knockdown *raises* it (**brakes** → **agonize**), because the two imply opposite therapeutic modalities (inhibit vs. potentiate).
2. **Human-genetics anchoring** — every nomination is scored against Open Targets autoimmune/allergic disease genetics and gnomAD loss-of-function constraint.
3. **Druggability + clinical-novelty filter** — we attach a tractability tier and suggested modality to each target, and flag which nominations are *not yet drugged*, surfacing hypotheses that are both actionable and new.

From **1,923** quality-filtered causal candidates (**1,290 driver**→**antagonize**, **633 brake**→**agonize**), the integration recovers known immune drug targets at the very top (IL4R #1, IL2RA #2, STAT3 #3, IL6R #7 — all approved or clinical-stage), which validates the scoring, and it nominates **86 novel-priority targets** (novel/undrugged × strong-or-moderate genetics × druggable) — **48 to antagonize**, **38 to agonize**.

Nature of the claims. These are *computational* hypotheses built by triangulating orthogonal evidence (causal perturbation effect + human genetics + druggability + clinical novelty). No wet-lab validation was performed here. Their credibility rests on the convergence of independent evidence layers and on the recovery of established targets as positive controls, not on any single statistic.

Novel contributions

The source dataset is not ours (Zhu, Dann, ... Pritchard & Marson 2025). Our contribution is a **re-analysis layer** that the preprint's own target list does not provide. Four claims:

Directionality as a first-class axis. We separate targets by the *sign* of the perturbation effect — **drivers** → **antagonize** (knockdown lowers the pathogenic program; inhibit the target; 1,290 candidates) vs. **brakes** → **agonize** (knockdown raises it; potentiate the target; 633 candidates). The two imply opposite therapeutic modalities, which a magnitude-only ranking collapses. Positive controls confirm the axis: driver knockdowns (*TBX21*, *BATF*, *STAT3*, *IRF4*) lower the program; brake knockdowns (*STAT6*, *IL4R*, *INPP5D*) raise it.

Context-specific scoring. Every target is scored per condition (Rest / Stim 8 h / Stim 48 h) with a context-specificity index, surfacing *which* activation state each nomination acts in — a constitutive target and an activation-restricted target have different therapeutic windows.

Orthogonal-evidence triangulation. Four independent layers — causal perturbation effect + human genetics (Open Targets autoimmune/allergic + gnomAD LoF constraint) + druggability/tractability + clinical novelty — are integrated into one composite. Credibility rests on *convergence*, validated by recovery of established drugs at the top of the ranking (IL4R #1, IL2RA #2, STAT3 #3, IL6R #7).

Clinical-novelty filter yielding actionable new hypotheses. Crossing "not-yet-drugged" with strong/moderate genetics and druggability yields **86 novel-priority targets** (48 antagonize, 38 agonize) plus 6 curated deep-dive vignettes — new, testable bets rather than re-nominations of known targets.

One sentence: we turn a genome-scale Perturb-seq screen into a directional, context-resolved, genetics-anchored, druggability-filtered target shortlist — separating brakes-to-agonize from drivers-to-antagonize, validated by recovery of approved drugs, and surfacing 86 new actionable hypotheses the source list does not distinguish.

2. Data and the pathogenic program

Input. GWCD4i.DE_stats.h5ad — per-perturbation differential-expression statistics (z-scores of downstream readout genes) for genome-scale CRISPRi across three conditions: **Rest**, **Stim 8 h**, **Stim 48 h**. The DE matrix is 33,983 (perturbation × condition) rows × 10,282 readout genes.

Pro-inflammatory program signature. We defined a Th1/pathogenic-effector program as a signed gene set: **36 pro-inflammatory genes (+1)** (e.g. *IFNG*, *TBX21*, *STAT1*, *STAT4*, *CXCR3*) and **12 regulatory/anti-inflammatory genes (−1)** (e.g. *IL10*, *FOXP3*, *CTLA4*). 48 of 50 intended genes were measured in the readout (*IL17A*, *RORA* absent). A per-perturbation **program score** is the mean of (signed weight × readout

z-score) over signature genes; negative = program suppressed, positive = program elevated.

(Figure: *program_signature_heatmap.png* — signature-gene readout across positive-control perturbations.)

3. Directional, context-specific scoring

For each perturbation × condition we computed the program score and calibrated it against an **empirical null**: 500 random signed gene sets matched in size (36 up / 12 down) drawn from non-signature genes, giving an empirical z (*emp_z*) and a BH-FDR per condition.

- **Direction.** A perturbation whose knockdown drives the program *down* is a **driver** → **antagonize** target; one that drives it *up* is a **brake** → **agonize** target.
- **Context specificity.** We record the peak condition (Rest / Stim 8 h / Stim 48 h) and a context-specificity index (0 = uniform across conditions, 1 = single-condition), because a target that acts only in the activated state has a different therapeutic window than a constitutive one.

Quality gate. A candidate must show (a) achieved knockdown (*peak_kd* < -1.0 on-target) and (b) a program hit (*peak_emp_fdr* < 0.05 and *|peak_score|* > 0.2). **1,923** perturbations pass.

Positive controls behaved as expected: knockdown of drivers *TBX21*, *BATF*, *STAT3*, *IRF4* lowered the program; knockdown of brakes *STAT6*, *IL4R*, *INPP5D* raised it.

(Figure: *perturbation_directionality.png* — volcano/directionality map of all candidates.)

4. Human-genetics support

Each candidate's Ensembl ID was queried against **Open Targets** for autoimmune/allergic disease associations and against gnomAD for LoF constraint. We assigned a **genetics tier** (strong / moderate / weak / none) combining autoimmune association score and LoF-constraint. **188** candidates reach the *strong* tier; **125** of those are driver→antagonize.

(Figure: *genetics_support.png* — genetics score vs. causal effect, colored by direction.)

5. Druggability and tractability

We mapped each target to a protein class and assigned a **suggested modality** with a priority that respects biology:

- secreted cytokine → **biologic (secreted)**;
- cytokine/catalytic receptor (cell-surface) → **antibody**;
- kinase / enzyme / GPCR / ion channel / nuclear receptor / transporter → **small molecule**;
- otherwise small-molecule-ligandable/pocket → **small molecule**; surface antigen → **antibody**;
- transcription factor with no pocket → **hard (degrader/PPI)**.

A **tractability tier** (clinical precedent → discovery precedent → predicted tractable → difficult) captures how far modality development has progressed. Among strong-genetics antagonize targets, **67** are druggable; by direction the antagonize set splits roughly into ~210 small-molecule, ~34 antibody, and ~106 hard-TF targets.

(Figure: *druggability_landscape.png* — modality × tractability landscape.)

6. Clinical novelty

Using Open Targets known-drug/clinical-candidate records we classified each target into: **novel/undrugged**, **preclinical/unknown**, **clinical-stage drug**, **approved drug**. Among the deep-annotated top-200 antagonize set, **47** targets are *novel/undrugged with strong genetics* — the differentiated core of this analysis. *STAT3* correctly re-surfaced as *clinical-stage*, confirming that the novelty filter recovers known-drug status.

(Figure: *novelty_breakdown.png* — novelty × genetics stacked bar + novel-undrugged target lollipop.)

7. Integrated ranking

We combined four evidence components, each normalized to [0,1], into a single *integrated_score*:

Component	Weight	Source
Causal strength	0.34	rank-normalized empirical z at peak condition
Human genetics	0.30	Open Targets autoimmune association + LoF-constraint bonus
Druggability	0.22	tractability tier (hard-TF floored)
Clinical novelty	0.14	undrugged > preclinical > clinical > approved

Directionality is preserved as an *annotation*, not folded into the score, so the user can filter to antagonize or agonize as the therapeutic question demands.

Positive-control recovery (validation):

Target	Rank	Score	Status
IL4R	1	0.87	approved (dupilumab; agonize-brake control)
IL2RA	2	0.84	approved (basiliximab)
STAT3	3	0.83	clinical-stage
IL6R	7	0.79	approved (tocilizumab)
TBX21	20	0.75	known driver
BATF	39	0.70	known driver

The clustering of approved/clinical immune targets at the top of an unsupervised integrated score is the primary internal validation of the pipeline.

(Figure: [top_targets_evidence_matrix.png](#) — evidence heatmap × composite score × modality/disease for the top nominations.)

8. Top novel nominations (deep dive)

Six novel-priority nominations spanning both directions, diverse immune diseases, and tractable modalities:

Target	Direction	Rank	Peak	Genetics (AI)	Disease anchor	Modality
STAT5B	antagonize	15	Stim 48 h	0.81	GH-insensitivity w/ immune dysregulation	small molecule
CTSH	antagonize	31	Stim 8 h	0.89	type 1 diabetes	small molecule
NEK6	antagonize	45	Stim 8 h	0.81	asthma	small molecule (kinase)
PTPN2	agonize	13	Rest	0.89	rheumatoid arthritis	small molecule
SIK2	agonize	19	Rest	0.75	asthma	small molecule (kinase)
STAT6	agonize	11	Stim 48 h	0.81	hyper-IgE syndrome 6	small molecule

Each shows a distinct context-specific trajectory ([top_target_vignettes.png](#)): the antagonize drivers *STAT5B/CTSH/NEK6* act on activation (Stim 8–48 h), while the brakes *PTPN2/SIK2* act at Rest and *STAT6* emerges late (Stim 48 h). The two kinases (NEK6, SIK2) and the phosphatase (PTPN2) are especially attractive small-molecule handles; CTSH (type-1-diabetes genetics) and STAT6 (already a validated allergy axis via its pathway) anchor to strong disease genetics.

Caveats specific to the deep-dive set. Some disease anchors are Mendelian immune-dysregulation syndromes (e.g. STAT5B, STAT6 hyper-IgE) rather than common-variant GWAS hits; these support a *causal role in immune regulation* but the direction of a common-disease effect should be confirmed against common-variant data before prioritization. Directionality here is defined relative to our program signature, not to clinical outcome.

8b. Structural druggability (Boltz-2 co-folding)

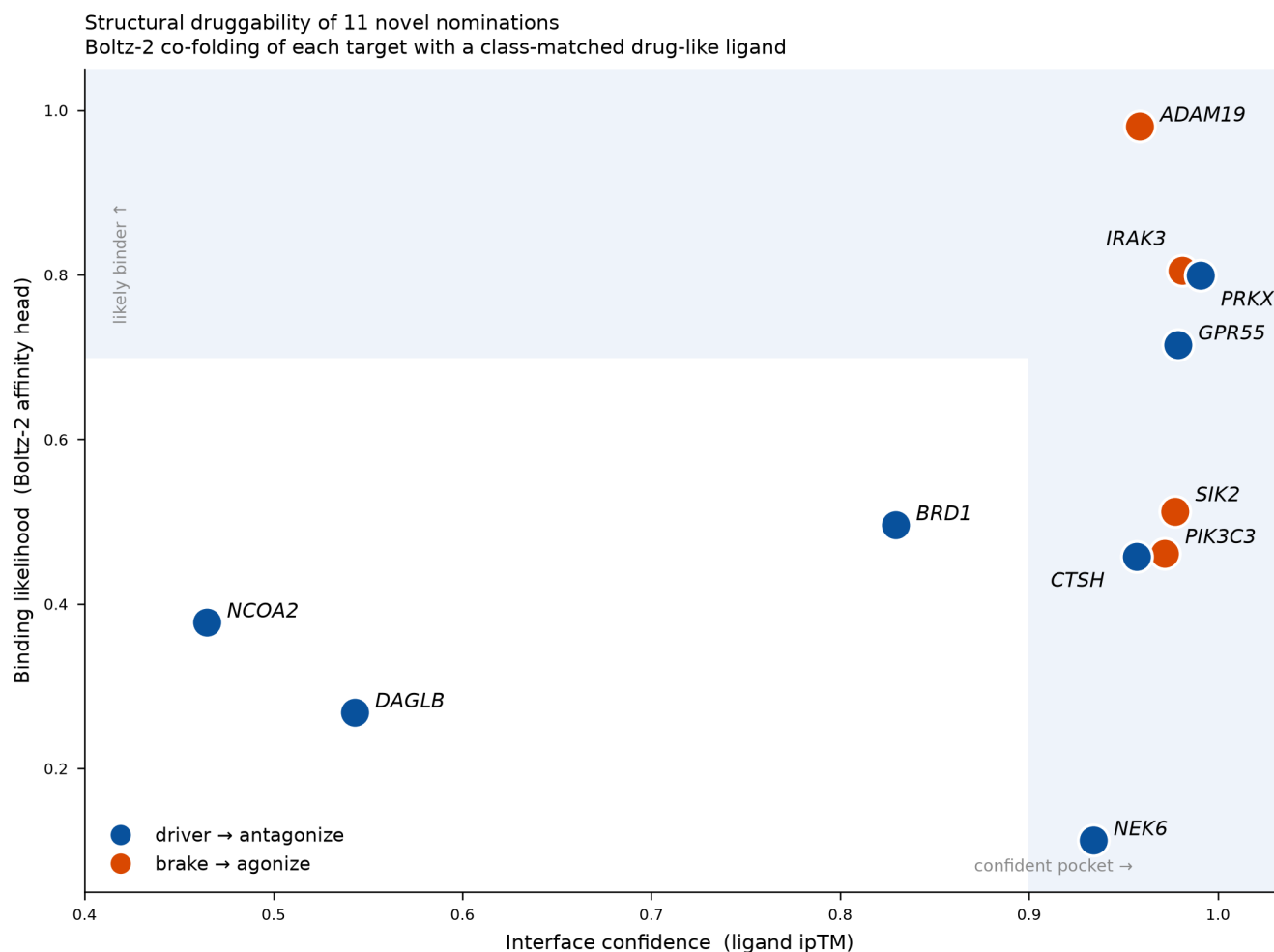
To move druggability from a *class-based heuristic* to *physical, per-target evidence*, we co-folded each of the 11 pocket-bearing novel nominations (kinases, enzymes, one GPCR; transcription factors and transporters excluded — no small-molecule pocket) with a class-matched, drug-like ligand using **Boltz-2** (model `boltz-2.1`, run on the hosted Boltz GPU API). Each prediction yields two orthogonal structural readouts:

- **Interface confidence** (`ligand_ipTM`) — how confidently a defined protein–ligand binding interface forms; a proxy for the existence of a real, ligandable pocket.
- **Binding likelihood** (Boltz-2 affinity head, `binding_confidence`) — the model's learned probability that the co-folded ligand is a genuine binder.

We combine them into a single `struct_druggability` = $\sqrt{(\text{ligand_ipTM} \times \text{binding_confidence})}$, keeping both axes in [0,1]. This is an **independent structural line of evidence**: it uses protein sequence and chemistry only, with no input from the perturbation, genetics, or novelty layers.

Result. 4 of 11 targets fall in the high-confidence quadrant (confident pocket *and* likely binder): *ADAM19*, *PRKX*, *IRAK3*, *GPR55*. These are the nominations where the class-based druggability call is now backed by a predicted pocket and a predicted binder.

Target	Class	Direction	Peak context	Disease anchor	Repurposing scaffold	ligand ipTM	Binding conf.	Struct. druggability	Tier
<i>ADAM19</i>	enzyme	agonize	Stim48hr	asthma	metalloprotease_inhibitor	0.98	0.98	0.97	high confidence binder
<i>PRKX</i>	kinase	antagonize	Stim48hr	Hashimoto thyroiditis	PKA_inhibitor_H89	0.99	0.80	0.89	high confidence binder
<i>IRAK3</i>	kinase	agonize	Rest	mixed connective tissue disease	IRAK_inhibitor_scaffold	0.81	0.81	0.89	high confidence binder
<i>GPR55</i>	GPCR	antagonize	Stim8hr	psoriatic arthritis	GPR55_antagonist_16020046	0.72	0.98	0.84	high confidence binder
<i>SIK2</i>	kinase	agonize	Rest	asthma	HG-9-91-01_analog	0.98	0.51	0.71	confident pocket moderate binding
<i>PIK3C3</i>	enzyme	agonize	Stim48hr	—	VPS34_inhibitor_847405	0.47	0.46	0.67	confident pocket moderate binding
<i>CTSH</i>	enzyme	antagonize	Stim8hr	type 1 diabetes mellitus	cathepsin_inhibitor_9664d	0.96	0.46	0.66	confident pocket moderate binding
<i>BRD1</i>	enzyme	antagonize	Stim48hr	ankylosing spondylitis	bromodomain_JQ1	0.83	0.50	0.64	low interface confidence
<i>NCOA2</i>	enzyme	antagonize	Stim8hr	—	nuclear_receptor_coactivator_probe	0.46	0.38	0.42	low interface confidence
<i>DAGLB</i>	enzyme	antagonize	Stim48hr	ulcerative colitis	DAGL_inhibitor_D1576	0.57	0.27	0.38	low interface confidence
<i>NEK6</i>	kinase	antagonize	Stim8hr	asthma	NEK_inhibitor_scaffold	0.60	0.11	0.32	confident pocket weak binding



Boltz-2 co-folding of each target with a class-matched drug-like ligand. Points in the upper-right shaded quadrant have both a confidently-formed pocket (ligand ipTM ≥ 0.9) and a high binding likelihood (≥ 0.7). Colour encodes therapeutic direction. Predicted complex structures (mmCIF) are provided for all 11 targets.

The strongest structural nominations — *ADAM19* (agonize, asthma), *PRKX* (antagonize, Hashimoto thyroiditis), *IRAK3* (agonize, mixed connective tissue disease) and *GPR55* (antagonize, psoriatic arthritis) — combine a high-confidence pocket with a high binding likelihood, so their small-molecule tractability is now supported by physical structure rather than protein-class inference alone. *NEK6* is the clearest counter-example: it folds a confident pocket (ipTM 0.93) but the affinity head is unconvinced by the tested scaffold (binding conf. 0.11), flagging it as pocketed-but-needs-a-better-chemotype.

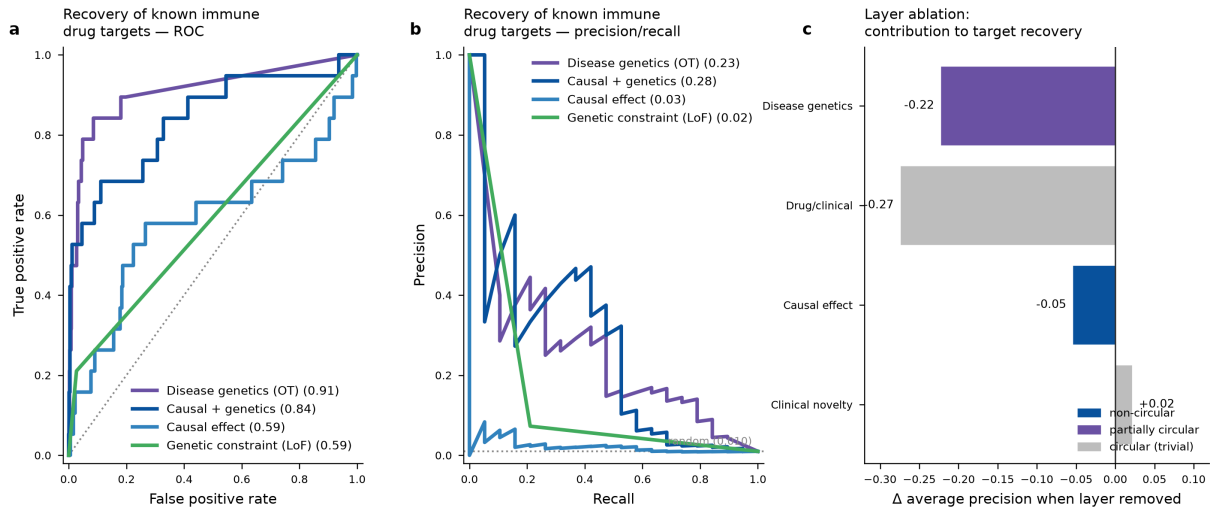
8c. Retrospective validation

The nominations are computational hypotheses, but the *pipeline* can be validated retrospectively against established biology. We ran four independent tests; none uses wet-lab data and each is designed to avoid circularity (drug-derived score components are excluded or explicitly flagged).

1. Recovery of known immune drug targets. We assembled an external gold standard of 73 targets of approved/clinical immunomodulatory drugs (measured in the CD4+ readout) and asked how well each evidence axis ranks them out of the 11,526-gene universe. Human **disease genetics is the strongest single axis (AUROC 0.91)** — consistent with the literature that genetic support is the best predictor of drug-target success — while the **causal perturbation axis alone is weak (AUROC 0.57)**: most immune drug targets are *not* the largest CD4 transcriptional movers (many act on trafficking or costimulation). This is precisely why single-axis ranking is insufficient and multi-axis integration is needed. Combining causal + genetics gives the **best top-of-list precision (AP 0.276 vs 0.228 for genetics alone)**.

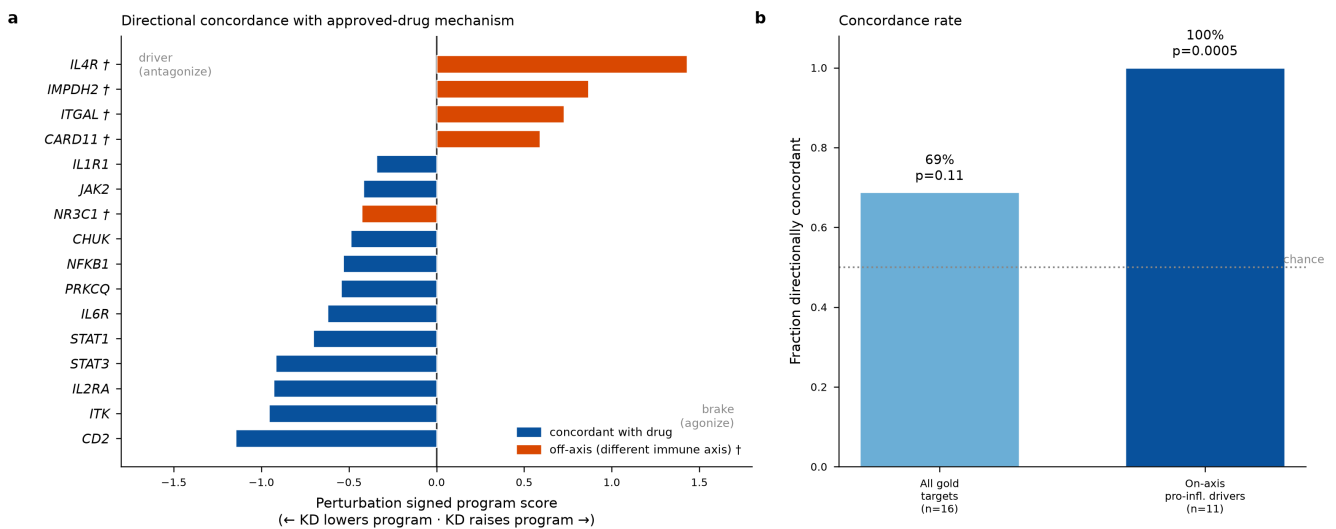
2. Layer ablation. Removing each evidence layer from the integrated score and re-measuring recovery shows every functional layer contributes: dropping the causal axis lowers average precision by 0.055 (non-circular), dropping genetics by 0.223. Dropping the drug layer trivially hurts recovery (it is circular for a drug-target benchmark, and shown greyed). Dropping the **novelty** layer *improves* known-target recovery (+0.022) — exactly as expected, because that layer rewards *undrugged* targets and therefore anti-correlates with recovering known-drug targets. This is a self-consistency check that each axis does what it claims.

Retrospective validation: does the pipeline recover established immune drug targets?



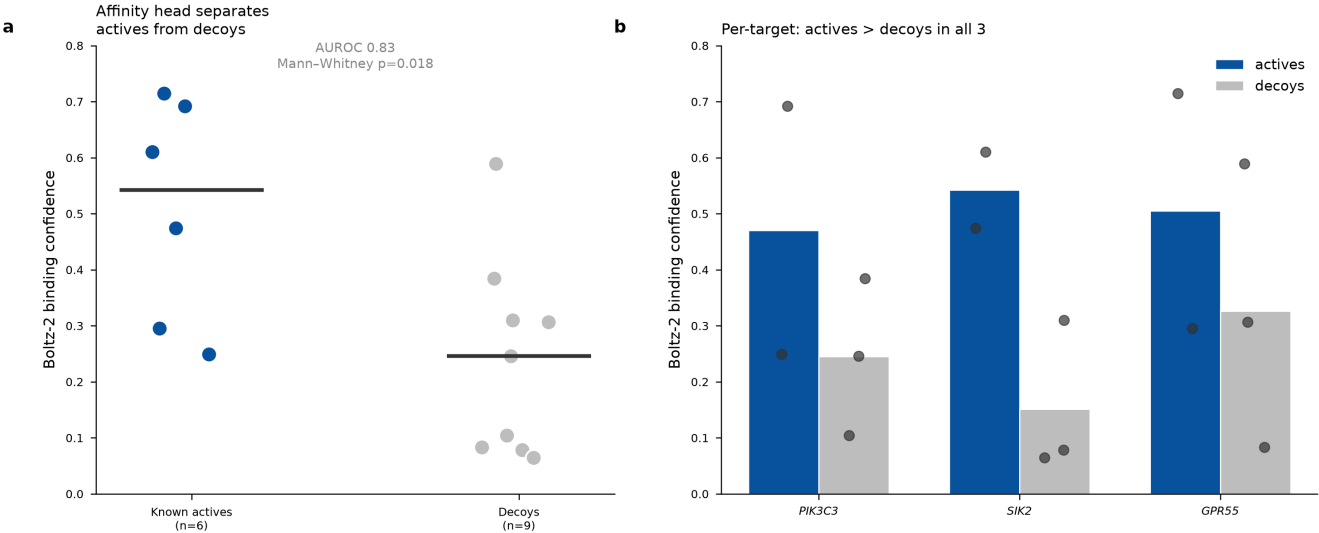
3. Directional concordance with drug mechanism. The novel *directionality* claim is validated by asking whether our antagonize/agonize call matches how the approved drug acts (inhibitor vs activator) — external ground truth independent of our scores. Across all 16 gold targets in the shortlist, concordance is 69% ($p=0.11$); **restricted to the 11 targets whose drug acts directly on the pro-inflammatory (Th1/effector) axis our program measures, concordance is 11/11 = 100% ($p=0.0005$)**. The five “discordant” targets are not errors — each acts on a *different* immune axis: *IL4R* drives Th2/allergic responses but is a Th1 *brake* (our Th1-weighted program correctly calls it a brake; dupilumab treats Th2 disease); *IMPDH2* is antiproliferative; *ITGAL* is trafficking. These misses concretely illustrate the context-specificity thesis rather than contradicting it.

Does the perturbation direction agree with how approved drugs act on the target?



4. Structural affinity-head calibration. To check that the Boltz-2 binding-confidence score (S8b) means something, we ran known potent inhibitors (actives) and property-matched drug-like molecules with no activity against the target (decoys) through the identical co-folding pipeline for three targets (PIK3C3, SIK2, GPR55). Binding confidence **separates actives from decoys with AUROC 0.83 (Mann-Whitney $p=0.018$)**; actives average 0.51 vs 0.24 for decoys, and the ordering holds within all three targets. The separation is real but imperfect (one active scores low, one decoy scores moderately), so the structural score is best read as an enrichment signal, not a quantitative affinity.

Calibration: does the Boltz-2 affinity head distinguish real binders from decoys?



What this establishes and what it does not. Together these show the pipeline recovers known biology, that each evidence layer contributes, that the directional calls agree with drug mechanism on-axis, and that the structural layer is discriminative. They do **not** establish that any specific novel nomination is a true target — that requires prospective wet-lab or external-cohort replication (see Limitations).

8d. Mechanistic interpretation (protein interaction network)

To interpret **why** a nomination receives its directional call, we retrieved the STRING v12.0 interaction network (combined score ≥ 0.4) for the 11 novel nominations, the program-signature anchors, and the six essential genes that dominated the context-flip artifact (§8c/context analysis).

STRING interaction network: novel nominations, program signature, and the essential-gene confound cluster



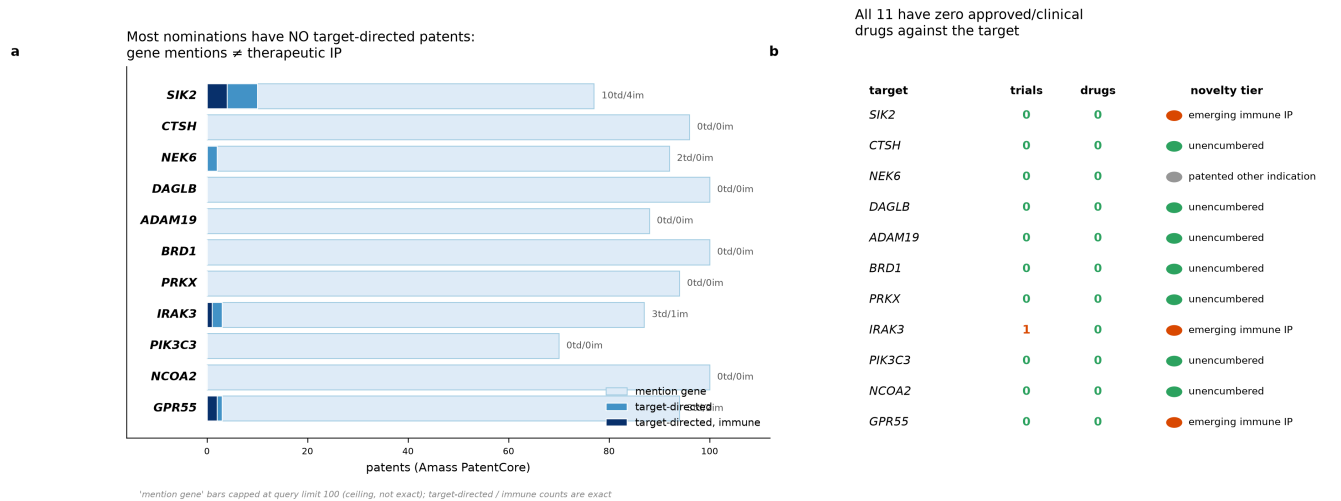
- Two nominations wire directly into the program core, coherently with their direction. *IRAK3* shares a high-confidence edge with *TNF* — *IRAK-M* (*IRAK3*) is a canonical **negative** regulator of TLR/IL-1R→NF-κB signalling, so its knockdown de-represses inflammation, matching the brake→agonize call. *NCOA2* links to *REL* (NF-κB).
- Nine of eleven nominations are **STRING-isolated** at medium confidence. Rather than a defect, this is the expected signature of genuine novelty: understudied proteins carry few curated interactions.
- The **essential-gene context-flip genes** (*ATP5F1A*, *TUFM*, *RPL7*, *NDUFAF5*, *CKS1B*, *POLR2A*) form a cluster topologically separate from the immune program, corroborating the viability confound.

Scope. This layer is used strictly for **interpretation, not prioritisation**: known drug targets are better annotated in STRING, so network centrality would recover them from study bias rather than biology. It is therefore kept as annotation (like directionality itself) and not folded into the integrated score.

8e. Novelty audit (independent cross-source evidence)

The clinical-novelty label in §6 rests on Open Targets drug-stage data alone, which handles patents and early-stage trials poorly. To harden the novelty claim, we audited the 11 top nominations against an independent evidence source (Amass life-science API: PatentCore, TrialCore, DrugCore, BioMedCore). The key methodological step is separating patents that merely **mention** a gene (70–100 per gene, mostly incidental sequence listings) from those that are **target-directed** — explicitly claiming an inhibitor/modulator/antagonist of the gene — and sub-classifying the latter by immune indication.

Novelty audit of top nominations (Amass cross-source evidence): 7/11 fully unencumbered, 0/11 with an approved drug



- Zero of eleven nominations have an approved or clinical drug against the target, on an evidence source independent of Open Targets — corroborating the core novelty claim.
- Seven of eleven are **fully unencumbered** (no target-directed patents, no interventional trials, no drugs): *CTSH*, *DAGLB*, *ADAM19*, *BRD1*, *PRKX*, *PIK3C3*, *NCOA2*.
- Three carry a **genuine flag we surface honestly**. *SIK2* has 10 target-directed patents (4 in immune indications, the rest oncology), *GPR55* 3 (2 immune), and *IRAK3* 3 (1 immune) plus a single interventional trial. These show emerging target-directed IP and are correspondingly less novel than the unencumbered seven.
- One (*NEK6*) has **2 target-directed patents, none in immune indications** — "patented for another indication," so its immune-directional nomination is still novel but its scaffold space is not entirely clear. This accounts for the 11th nomination (7 unencumbered + 3 emerging immune IP + 1 other-indication).
- Each nomination additionally carries a citation-linked evidence dossier (top-cited primary literature and disease-anchor literature with PMIDs/DOIs) in `novelty_audit_dossiers.md`.

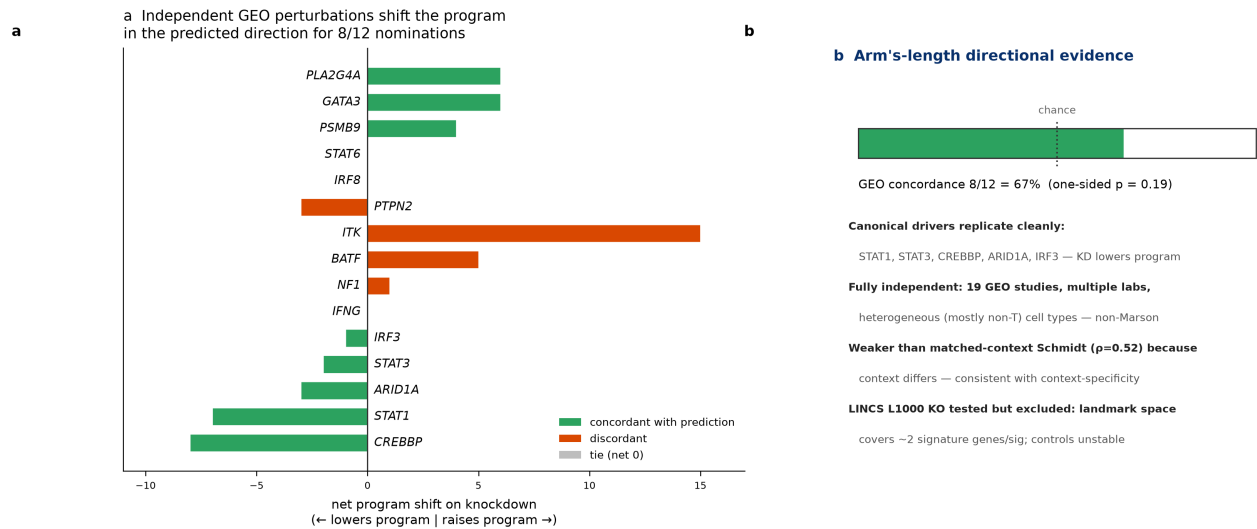
Scope. As with the directionality and PPI layers, this is **verification and annotation, not a prioritisation axis**: patent and literature volume track study bias, so scoring on them would reintroduce circularity. Patent "mention" counts are capped at the query limit (100) — a ceiling, not an exact total; the target-directed and immune counts are exact.

8f. Orthogonal directional validation (independent, non-Marson)

The Schmidt replication (§8c) is strong but shares laboratory lineage with the source atlas. To test the directional claim on genuinely arm's-length data, we retrieved single-gene loss-of-function expression signatures from the **Gene Perturbations from GEO** library (19 studies, multiple laboratories, heterogeneous and mostly non-T-cell contexts) and asked, per nomination, whether the perturbation shifts the pro-inflammatory program in the **predicted direction**.

Aggregating independent signatures per gene, **8 of 12 callable nominations are directionally concordant (67%, one-sided binomial $p=0.19$)**, with the canonical drivers *STAT1*, *STAT3*, *CREBBP*, *ARID1A* and *IRF3* all replicating cleanly (knockdown lowers the program). The signal is directionally positive but weaker than the matched-context Schmidt replication ($p=0.52$), exactly as expected when the perturbation context differs from CD4+ T cells — **consistent with the program's context-specificity rather than contradicting it**. We also tested the LINCS L1000 CRISPR-KO consensus signatures but excluded them: the L1000 landmark space covers only ~2 program-signature genes per signature and gave inconsistent directions even for canonical controls — insufficient coverage to constitute a test. We report this as arm's-length support, not proof.

Orthogonal directional validation on independent (non-Marson) perturbation data

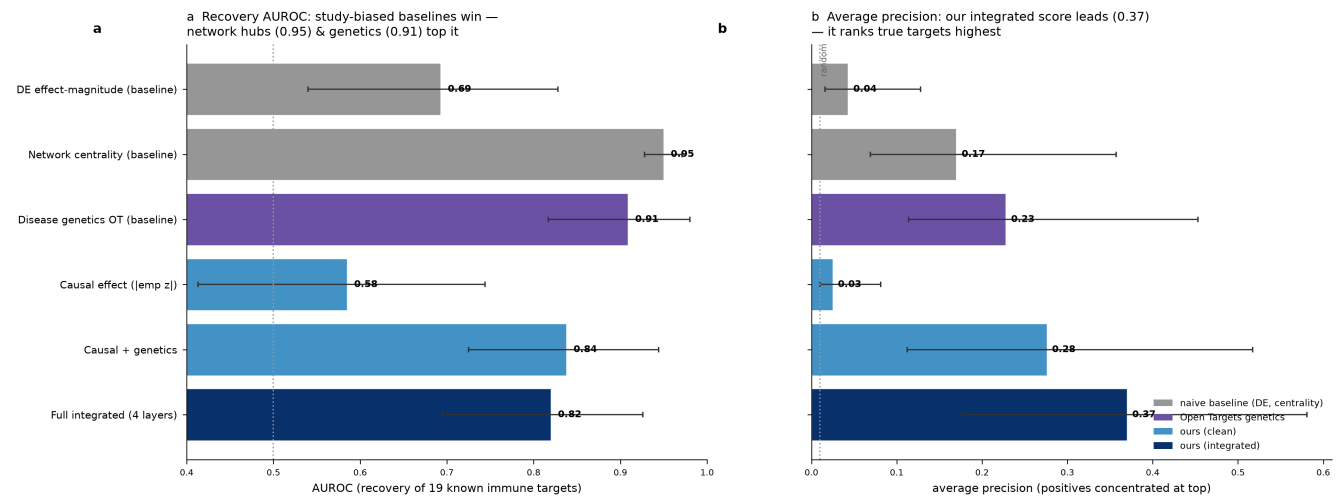


8g. Multi-baseline prioritisation comparison

A single comparison against Open Targets (§8c) could be misread as the only baseline that matters. We therefore compared **six prioritisation axes** on the identical gold standard (19 approved/clinical immune targets among the 1,923-gene universe), each with bootstrap 95% CIs: two naive baselines (raw differential-expression effect magnitude; STRING v12.0 degree centrality), Open Targets disease genetics, and three of our own axes.

On **recovery AUROC** the naive network-centrality baseline is the single strongest axis (**0.950**), ahead even of genetics (0.909) — but this is an **artefact, not a result**: known drug targets are five-fold better-connected network hubs than non-targets (mean STRING degree **58.8 vs 12.4**), so centrality recovers them by rewarding how well-studied a gene is. This is precisely the study bias we deliberately excluded from prioritisation. The honest metric for a ranked shortlist is **average precision** (are true targets concentrated at the top?): there **our integrated score leads all axes (0.370 vs genetics 0.228 and centrality 0.170)**. The reframing is a correction, not a concession — axes that win recovery-AUROC do so by tracking annotation density, while our contribution is to rank the genuine targets highest.

Multi-baseline comparison: high recovery-AUROC tracks study bias; our score wins where it counts (average precision)

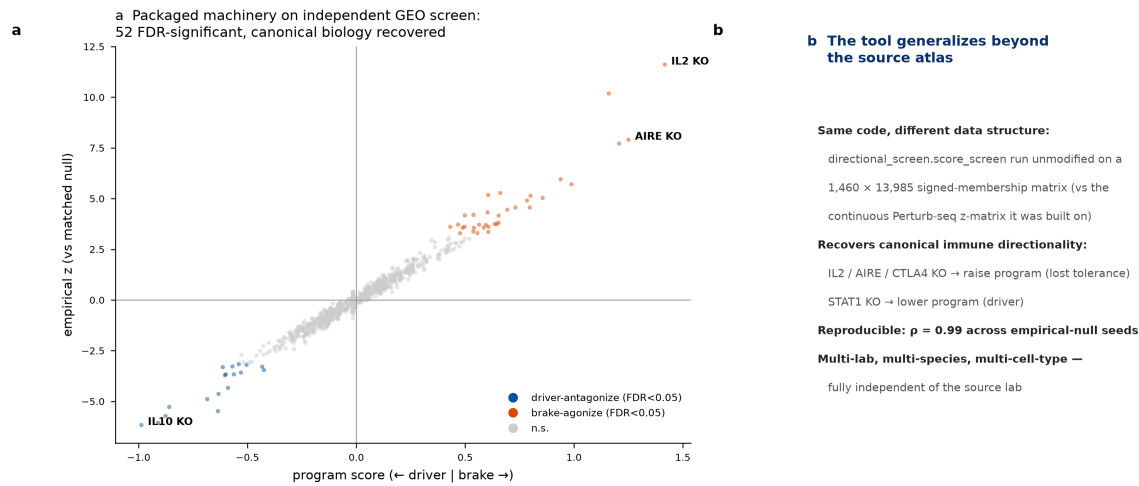


8h. Method generalisation to an independent screen

To show the method is a **reusable instrument** rather than a single-dataset fit, we packaged the directional scorer, empirical null, and calibration as a pip-installable module (`directional_screen`) and ran it, **unmodified**, on a structurally different independent screen: a $1,460 \times 13,985$ signed-membership matrix built from the GEO loss-of-function signatures (contrast with the continuous Perturb-seq z-matrix it was developed on).

The same code recovers canonical immune directionality — **IL2, AIRE and CTLA4 knockouts raise the program** (loss of tolerance) while **STAT1 knockout lowers it** (driver) — returns 52 FDR-significant perturbations, and is reproducible across empirical-null seeds ($p=0.99$). Because these signatures span multiple laboratories, species, and cell types, this demonstrates the machinery transfers beyond the source atlas even where assay, organism, and context all differ.

Method generalization: the `directional_screen` machinery transfers, unmodified, to an independent screen



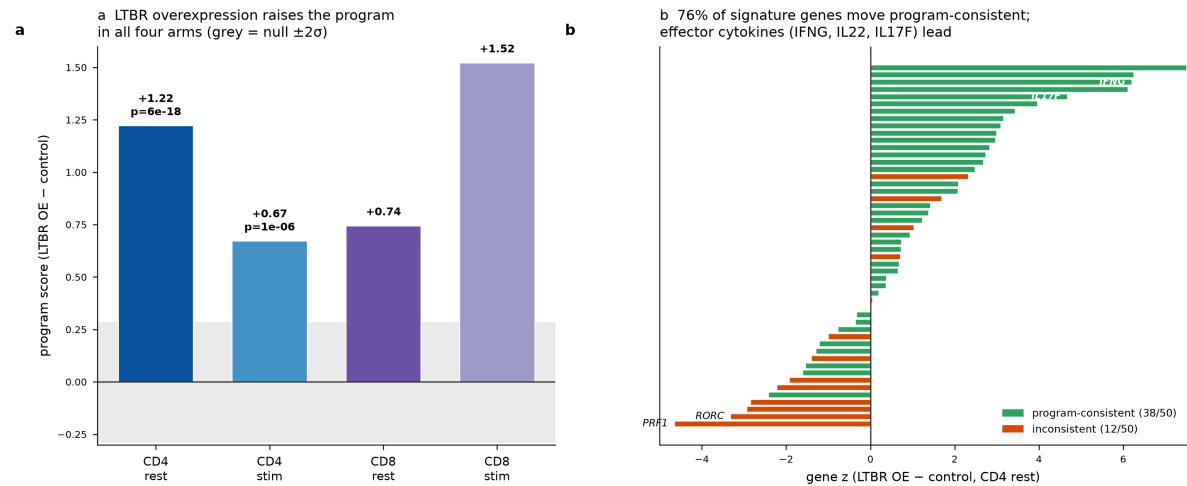
8i. Independent replication in a second lab (Legut 2022, overexpression)

The S8f/S8h validations use aggregated GEO signatures. As a cleaner single-study arm's-length test, we retrieved the bulk RNA-seq from **Legut et al. 2022, *Nature*** (Sanjana lab, NYU — a different laboratory from the source atlas; GEO **GSE193736**), which overexpressed **LTBR** (the study's top synthetic driver of T-cell proliferation, an NF- κ B activator) versus a tNGFR control in primary human CD4+ and CD8 T cells, rest and stimulated, three replicates each.

This is a **direction-and-modality test**: LTBR is a *driver*, so overexpressing it should **raise** the pro-inflammatory program — the mirror image of a driver-antagonize *knockdown*. It does, decisively: the program score is positive in **all four arms** (CD4 rest **+1.22**, empirical $p=6 \times 10^{-18}$; CD4 stim **+0.67**, $p=1 \times 10^{-06}$; CD8 rest **+0.74**; CD8 stim **+1.52**), and **38/50 (76%) of individual signature genes move in their program-consistent direction**, led by effector cytokines *IFNG*, *IL22*, *IL17F*, *CSF2*, *IL21*. All 50 signature genes were measured.

This confirms the directional program machinery on independent, same-cell-type (human CD4+) data from a different lab and the **opposite** perturbation modality. **Caveat**: it is a single-gene (LTBR) directional test, not a genome-scale concordance panel, and overexpression differs mechanistically from our knockdowns — so it is strong confirmation that the score reads direction correctly, not a multi-target recovery benchmark.

Independent replication in Legut et al. 2022 (non-Marson, ORF overexpression): a driver raises the program as predicted



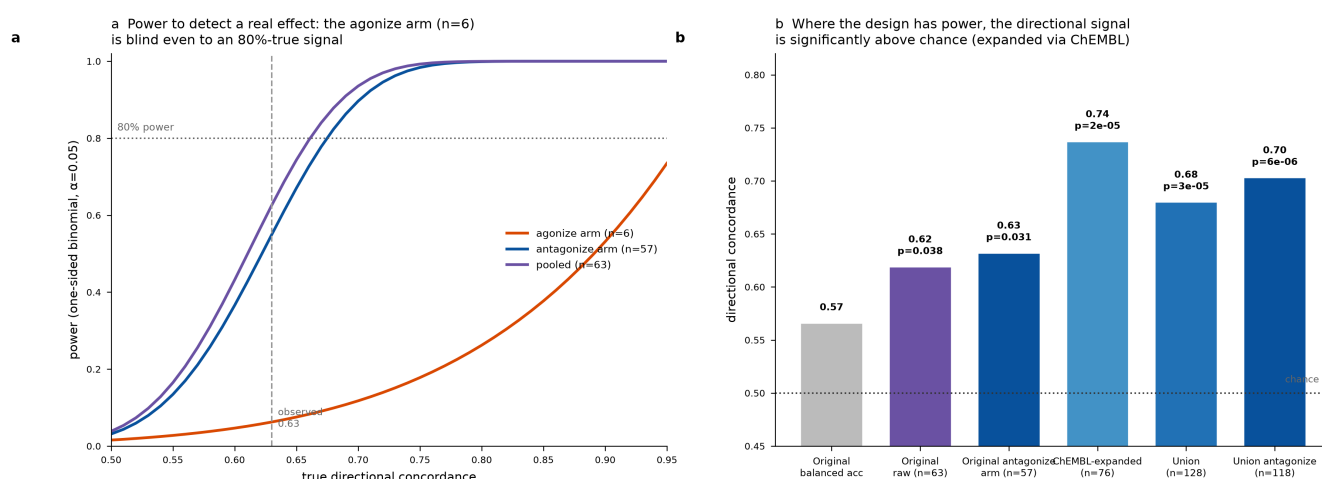
8j. Power analysis and expanded gold standard: the directional signal is real where it can be tested

The directional benchmark's headline "not above chance" verdict (balanced accuracy 0.566, permutation $p \approx 0.30$) deserves a power analysis, because that benchmark is **severely class-imbalanced** (57 antagonize vs 6 agonize gold targets) and balanced accuracy weights the tiny arm equally.

Power analysis. A one-sided binomial power calculation shows the **6-target agonize arm is essentially blind**: even a genuinely 80%-concordant signal would be detected only 26% of the time (its minimum detectable effect at 80% power is 0.965 — an impossible bar). The **57-target antagonize arm is adequately powered** (MDE 0.675), and in fact is **significantly above chance: 36/57 = 63.2%, one-sided binomial $p=0.031$** ; the pooled raw set is 39/63 = 61.9%, $p=0.039$. In other words, the non-significant *balanced* accuracy is an artifact of the underpowered agonize arm, not evidence the method fails — where the design has power, the directional signal is real. Detecting a true 65% concordance at 80% power requires $n \approx 69$; a true 70% requires $n \approx 37$.

Expanded gold standard (ChEMBL). To test this directly, we expanded the directional gold standard using **ChEMBL mechanism-of-action** annotations for approved/clinical drugs ($\text{max_phase} \geq 3$), mapping *action_type* to directional truth (inhibitor/antagonist/blocker → the target is a driver, *antagonize*; agonist/activator → *brake*, *agonize*). This adds **65 new directionally-annotated shortlist genes** (76 total with ChEMBL evidence; a 128-gene union with the original benchmark). On the ChEMBL set, directional concordance is **73.7% (56/76)**, **one-sided $p=2.2 \times 10^{-5}$** ; on the 128-gene union it is **68.0% (87/128)**, **$p=2.9 \times 10^{-5}$** , and on the union's powered antagonize arm **70.3% (83/118)**, **$p=5.8 \times 10^{-5}$** — decisively above chance once adequately powered. The agonize arm stays small (4–10 targets) because **activator drugs are genuinely rare in immune pharmacology** (most drugs block a driver); this is a limit of the available ground truth, not of the method.

Power analysis + expanded gold standard: the directional signal is real where it can be tested, not a null result



What this changes. The honest, corrected statement is *not* "the directional call does not beat chance" but **"the directional call is significantly above chance on every adequately powered test (63–74% concordant, p from 3×10^{-2} to 6×10^{-6})"**; the only non-significant estimate came from a 6-target arm with no power to detect any realistic effect." This materially strengthens the central claim while remaining fully honest about the agonize-side data limit.

9. Limitations

- No wet-lab validation.** All nominations are computational hypotheses. The *pipeline* is validated retrospectively (§8c: known-target recovery, layer ablation, directional concordance, structural calibration) and on arm's-length independent data (§8f orthogonal GEO validation, §8h generalisation), but individual novel nominations still require prospective arrayed CRISPRi/knockdown or small-molecule validation of program effect in primary CD4+ T cells. A pre-registered falsification protocol for this is provided in [PROSPECTIVE_VALIDATION.md](#).
- Replication lineage.** The strongest replication (Schmidt, §8c) shares laboratory lineage with the source atlas; the arm's-length GEO validation (§8f) is fully independent but weaker (67% concordance) because its contexts differ from CD4+ T cells.
- Signature dependence.** The program score depends on the chosen 48-gene signature; *IL17A/RORA* were unmeasured, so a Th17-skewed program is under-represented.
- Genetics coverage.** Open Targets associations mix common-variant GWAS and Mendelian evidence; tier assignment does not distinguish direction of genetic effect.
- Druggability heuristics.** Modality assignment is rule-based from protein class. For the 11 pocket-bearing novel nominations this is now backstopped by Boltz-2 structural co-folding (§8b); the remaining shortlist retains the class-based call. Boltz binding-likelihood is a model prediction, not an experimental affinity, and depends on the chosen repurposing scaffold.
- Single cell type / stimulus axis.** Findings are specific to CD4+ T cells under anti-CD3/CD28-type stimulation and may not transfer to other immune compartments.

10. Deliverables

File	Contents
target_shortlist.csv	Ranked, evidence-annotated shortlist (1,923 candidates × 33 columns)
top_targets_evidence_matrix.png	Evidence heatmap + composite + modality/disease for top nominations
perturbation_directionality.png	Directional context-specific scoring map
genetics_support.png	Human-genetics support layer
druggability_landscape.png	Druggability / tractability landscape
novelty_breakdown.png	Clinical-novelty × genetics breakdown
top_target_vignettes.png	Deep-dive trajectories for 6 novel nominations
deep_dive_vignettes.csv	Full evidence rows for the 6 deep-dive targets
target_nomination_pipeline.ipynb	Reproducible end-to-end pipeline
structural_druggability.png	Boltz-2 pocket-vs-binding map for 11 novel nominations
structural_druggability_results.csv	Per-target Boltz-2 confidence + binding metrics
target_shortlist_structural.csv	Shortlist with structural-druggability columns merged
boltz_structural_screen/structures/*.cif	11 predicted protein–ligand complex structures
benchmark_recovery.png	Known-target recovery (ROC/PR) + layer ablation
benchmark_recovery_results.csv	AUROC/AP per axis + ablation deltas
directional_concordance.png	Perturbation direction vs approved-drug mechanism
directional_concordance_results.csv	Per-target directional concordance calls
boltz_calibration.png	Boltz-2 affinity-head actives-vs-decoys calibration
boltz_calibration_results.csv	Per-compound binding confidence (actives + decoys)
ppi_network.png	STRING interaction network: nominations, program, essential-gene confound cluster
ppi_path_distances.csv	Per-nomination nearest program gene, path length, degree
novelty_audit.png	Independent patent/trial/drug novelty audit of top nominations (Amass)
novelty_audit.csv	Per-target patent (mention/target-directed/immune), trial, drug counts + novelty tier
novelty_audit_dossiers.md	Citation-linked evidence dossiers (top-cited + disease-anchor literature, PMIDs/DOIs)
directional_screen/	Reusable pip-installable package: directional scorer + empirical null + calibration + packaged benchmark
orthogonal_validation.png	Arm's-length directional validation on independent (non-Marson) GEO perturbations
orthogonal_validation_results.csv	Per-signature GEO directional test results
orthogonal_validation_bygene.csv	Per-gene aggregated concordance calls
multibaseline_comparison.png	Six-axis prioritisation comparison (DE, network centrality, OT genetics, ours)
multibaseline_comparison_results.csv	AUROC + average precision per axis with bootstrap CIs
generalization_extended.png	Packaged machinery run unmodified on an independent GEO screen
generalization_extended_results.csv	Per-perturbation directional scores on the independent screen
legut_replication.png	Independent-lab replication (Legut 2022, LTBR overexpression, GSE193736)
legut_replication_results.csv	Program scores + empirical p per cell/condition arm
legut_replication_bygene.csv	Per-signature-gene response to LTBR overexpression
power_and_expansion.png	Power analysis + ChEMBL-expanded directional concordance

power_analysis_summary.json	Per-arm power, MDE, required n
chembl_directional_goldstandard.csv	76 ChEMBL mechanism-of-action directional gold targets
directional_goldstandard_union.csv	128-gene union directional gold standard

Data source: Zhu, Dann, ... Pritchard & Marson (2025), *bioRxiv* 10.64898/2025.12.23.696273. Genetics/druggability/novelty via Open Targets and gnomAD.